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Differential Calculus

Standard Results

① $\frac{d}{dx} x^n = n x^{n-1}$

② $\frac{d}{dx} (\log x) = \frac{1}{x}$

③ $\frac{d}{dx} e^{ax} = a e^{ax}$

④ $\frac{d}{dx} k = 0$ where k is const.

⑤ $\frac{d}{dx} k \cdot f(x) = k \cdot \frac{df(x)}{dx} = k f'(x)$

⑥ $\frac{d}{dx} \sin x = \cos x$

⑦ $\frac{d}{dx} \cos x = -\sin x$

⑧ $\frac{d}{dx} \tan x = \sec^2 x$

⑨ $\frac{d}{dx} \cot x = -\operatorname{cosec}^2 x$

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(10) $\frac{d}{dx} \sec x = \sec x \cdot \tan x$

(11) $\frac{d}{dx} \operatorname{cosec} x = -\operatorname{cosec} x \cdot \cot x$

(12) $\frac{d}{dx} (a^x) = a^x \cdot \log_e(a)$

(13) $\frac{d}{dx} (\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

(14) $\frac{d}{dx} (\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$

(15) $\frac{d}{dx} (\tan^{-1} x) = \frac{1}{1+x^2}$

(16) $\frac{d}{dx} (\cot^{-1} x) = \frac{-1}{1+x^2}$

(17) $\frac{d}{dx} (\sec^{-1} x) = \frac{1}{x\sqrt{x^2-1}}$

(18) $\frac{d}{dx} (\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$

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★ Linear property of Derivative

$$\frac{d}{dx}\left[a f(x) + b g(x) - c h(x)\right] = a f'(x) + b g'(x) - c h'(x)$$

★ product rule of Derivative

$$\frac{d}{dx}(u \cdot v) = u \frac{dv}{dx} + v \cdot \frac{du}{dx}$$

★ Division Rule of Derivative

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{\left[v \frac{du}{dx} - u \frac{dv}{dx}\right]}{v^2}$$

★ Chain Rule for derivative (Derivatives of parametric eqns)

If $x = f(t)$, $y = g(t)$ where t is parameter

Then. $$\frac{dy}{dx} = \frac{dy}{dt} \times \frac{dt}{dx}$$

$$= \frac{dy/dt}{dx/dt}$$

Hyperbolic function

(1) $\sinh(x) = \frac{e^x - e^{-x}}{2}$

(2) $\cosh(x) = \frac{e^x + e^{-x}}{2}$

(3) $\tanh(x) = \frac{e^x - e^{-x}}{e^x + e^{-x}}$

(4) $\coth(x) = \frac{e^x + e^{-x}}{e^x - e^{-x}}$

(5) $\operatorname{sech}(x) = \frac{2}{e^x + e^{-x}}$

(6) $\operatorname{cosech}(x) = \frac{2}{e^x - e^{-x}}$

★ **Derivatives of Hyperbolic function :⇒**

(1) $\frac{d}{dx}\sinh(x) = \cosh x$ | (2) $\frac{d}{dx}\cosh x = \sinh x$

(3) $\frac{d}{dx}\tanh(x) = \operatorname{sech}^2(x)$ | (4) $\frac{d}{dx}\coth(x) = -\operatorname{cosech}^2 x$

(5) $\frac{d}{dx}\operatorname{sech}(x) = -\operatorname{sech} x \cdot \tanh x$ | (6) $\frac{d}{dx}\operatorname{cosech}(x) = -\operatorname{cosech} x \cdot \coth x$

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Derivative of Inverse Hyperbolic function :

(1) $\frac{d}{dx} \sinh^{-1}(x) = \frac{1}{\sqrt{1+x^2}}$

(2) $\frac{d}{dx} \cosh^{-1}(x) = \frac{1}{\sqrt{x^2-1}}$

(3) $\frac{d}{dx} \tanh^{-1}(x) = \frac{1}{1-x^2}$ where $|x|<1$

(4) $\frac{d}{dx} \coth^{-1}(x) = \frac{-1}{x^2-1}$ where $|x|>1$

(5) $\frac{d}{dx} \operatorname{sech}^{-1}(x) = \frac{-1}{x\sqrt{1-x^2}}$

(6) $\frac{d}{dx} \operatorname{cosech}^{-1}(x) = \frac{-1}{x\sqrt{x^2+1}}$

Remember

(1) $\cosh^2(x) - \sinh^2(x) = 1$

(2) $\operatorname{sech}^2(x) = 1 - \tanh^2 x$

(3) $\operatorname{cosech}^2(x) = -1 + \coth^2 x$

(4) $\sinh^{-1}(x) = \log\left[x + \sqrt{x^2+1}\right]$

(5) $\cosh^{-1}(x) = \log\left[x + \sqrt{x^2-1}\right]$

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★ Successive Differentiation

Let $y = f(x)$ be the any curve (equation).

Then.

Derivative of $f(x)$ w.r.to. x is ~~given by~~ denoted by

$$\frac{dy}{dx} \text{ or } Dy \text{ OR } f'(x) \text{ OR } y' \text{ or } y_1$$

called as first differentiation of $y = f(x)$.

If $\frac{dy}{dx}$ is further differentiated w.r.to x we get

$$\frac{d}{dx}\left(\frac{dy}{dx}\right) = \frac{d^2y}{dx^2} \text{ OR } D^2y \text{ OR } f''(x) \text{ OR } y'' \text{ or } y_2$$

called as second derivative of $f(x) = y$.

Similarly,

$$\frac{d^3y}{dx^3}, \text{ OR } D^3y \text{ OR } f'''(x) \text{ OR } y''' \text{ OR } y_3$$

$$\frac{d^4y}{dx^4} \text{ OR } D^4y \text{ OR } f^{IV}(x) \text{ OR } y^{IV} \text{ OR } y_4$$

$$\vdots \qquad \vdots \qquad \vdots \qquad \vdots$$

$$\frac{d^ny}{dx^n} \text{ OR } D^ny \text{ OR } f^n(x) \text{ OR } y^n \text{ OR } y_n$$

This represent the successive differentiation of function $y = f(x)$

Here. $\frac{d^ny}{dx^n}$ OR D^ny or y_n represent n^{th} derivation of $y = f(x)$ w.r.to x.

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★ n^{th} Derivative of some standard functions

(1) If $y = e^{(ax+b)}$

$\Rightarrow$ Diff. w.r.to x. we get

$$y_1 = \frac{dy}{dx} = a \cdot e^{(ax+b)}$$

again diff. w.r.to x. successively

$$y_2 = \frac{d^2y}{dx^2} = a^2 e^{(ax+b)}$$

$$y_3 = \frac{d^3y}{dx^3} = a^3 e^{(ax+b)}$$

⋮

In general

$$\boxed{y_n = \frac{d^ny}{dx^n} = a^n \cdot e^{(ax+b)}}$$

Case (i): If $a=1$ & $b=0$ i.e. $\boxed{y = e^x}$

Then $\boxed{y_n = (1)^n \cdot e^{ax} = e^x}$

Case (ii): If $b=0$, i.e. $y = e^{ax}$

Then. $\boxed{y_n = a^n e^{ax}}$

Case (iii):

If $a=-1$, $b=0$, i.e. $y = e^{-x} = 1/e^x$

Then $\boxed{y_n = (-1)^n e^{-x}}$

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(2) Let $y = (ax+b)^m$ where $m = 1, 2, 3, \ldots \in N$

Diff. w.r to x. we get

$$y_1 = m \cdot a (ax+b)^{m-1}$$

again diff. w.r to x. successively.

$$y_2 = m \cdot (m-1) \cdot a^2 (ax+b)^{m-2}$$

$$y_3 = m(m-1)(m-2)\, a^3 (ax+b)^{m-3}$$

$$y_4 = m(m-1)(m-2)(m-3) \cdot a^4 (ax+b)^{m-4}$$

$$\vdots$$

In general

$$y_n = m(m-1)(m-2)\ldots\ldots(m-n+1)\, a^n \cdot (ax+b)^{m-n}$$

$$= \frac{[m(m-1)(m-2)\ldots(m-n+1)][(m-n)(m-n-1)\ldots 3.2.1]\, a^n \cdot (ax+b)^{m-n}}{(m-n)(m-n-1)\ldots 3.2.1}$$

$D^n (ax+b)^m \equiv \dfrac{d^n (ax+b)^m}{dx^n}$

$$\boxed{y_n = \frac{m!}{(m-n)!}\, a^n \cdot (ax+b)^{m-n}}, \quad \underline{m \geq n}$$

Case (I): If $\underline{m=n}$ then $y_n = D^n (ax+b)^n = n!\, a^n$

Case (II): If $\underline{m<n}$ then $y_n = 0$

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(3) If $y = (ax+b)^{-1} = \frac{1}{(ax+b)}$

Diff. w.r.to x. we get

$$y_1 = \frac{-1}{(ax+b)^2} \cdot a$$

again Diff w.r.to. x, successively. we get

$$y_2 = \frac{(-1)(-2)}{(ax+b)^3} \cdot a^2 = \frac{(-1)^2 \cdot 2 \cdot a^2}{(ax+b)^3}$$

$$y_3 = \frac{(-1)^2 \cdot (-3) \cdot 2 \cdot a^3}{(ax+b)^4} = \frac{(-1)^3 \cdot (3\times 2) \cdot a^3}{(ax+b)^4}$$

$$y_4 = \frac{(-1)^3 \ -4\times 3\times 2 \cdot a^4}{(ax+b)^5} = \frac{(-1)^4 \ 4\times 3\times 2 \cdot a^4}{(ax+b)^5}$$

$$= \frac{(-1)^4 \cdot 4! \ a^4}{(ax+b)^5}$$

⋮

In general

$$\boxed{y_n = \frac{(-1)^n \cdot n! \ a^n}{(ax+b)^{n+1}}}$$

Similarly: If $y = (ax+b)^{-m}$

$y_n =$

$$\boxed{y_n = \frac{(-1)^n \ [m+(n-1)]! \ a^n}{(m-1)! \ (ax+b)^{m+n}}}$$

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(9) Let $y = a^{mx}$

Diff. w.r.to x. we get

$$y_1 = m \cdot a^{mx} \cdot \log a$$

again Diff. w.r.to x, successively.

$$y_2 = m^2 \, a^{mx} \cdot (\log a)^2$$

$$y_3 = m^3 \, a^{mx} \cdot (\log a)^3$$

$$\vdots$$

$$y_n = m^n \, a^{mx} \cdot (\log a)^n$$

Case (1): If $m = 1$ $\boxed{y = a^x}$

then $\because y_n = (1)^n \cdot a^x (\log a)^n$

$$\Rightarrow \boxed{y_n = a^x (\log a)^n}$$

---

let $y = (ax+b)^{-m}$

$\Rightarrow y_1 = -m \, (ax+b)^{-m-1} \cdot a$

$y_2 = -m(-m-1) \cdot (ax+b)^{-m-2} \cdot a^2$

$y_3 = -m \cdot (-m-1)(-m-2) \cdot (ax+b)^{-m-3} \cdot a^3$

$= \dfrac{(-1)^3 (m)(m+1)(m+2) \, a^3}{(ax+b)^{m+3}}$

$= \dfrac{(-1)^3 (m+2)(m+1)(m) \cdot a^3}{(ax+b)^{m+3}} \times \dfrac{(m+1)!}{(m+1)!}$ $\quad \therefore y_3 =$

Thus in general $\leftarrow$ $= \dfrac{(-1)^3 (m+2)! \, a^3}{(m-1)! \, (ax+b)^{m+n}} = \dfrac{(-1)^3 (m+(3-1))! \, a^3}{(m-1)! \, (ax+b)^{m+3}}$

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⑤ let $y = \sin(ax+b)$

Diff. w.r.to x. we get

$$y_1 = a \cdot \cos(ax+b) = a \sin(ax+b+\pi/2)$$

$\because \sin\theta = \cos(\theta+\pi/2)$

again diff. w.r.to. x., successively

$$y_2 = a^2 \cos(ax+b+\pi/2) = a^2 \sin[ax+b+2(\pi/2)]$$

$$y_3 = a^3 \cos(ax+b+2\pi/2) = a^3 \sin(ax+b+3\pi/2)$$

⋮

in general

$$\boxed{y_n = a^n \cdot \sin(ax+b+n\pi/2)}$$

⑥ let $y = \cos(ax+b)$

$\because -\sin\theta = \cos(\theta+\pi/2)$

Then

$$\boxed{y_n = a^n \cos\left(ax+b+n\frac{\pi}{2}\right)}$$

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(7) let $y = e^{ax} \cdot \sin(bx+c)$

Diff. w.r.to. x. we get

$$y_1 = a e^{ax} \cdot \sin(bx+c) + b e^{ax} \cdot \cos(bx+c)$$

$$= e^{ax} [a \sin(bx+c) + b \cos(bx+c)] \quad \text{---(1)}$$

Now put $a = r\cos\theta$, $b = r\sin\theta$

$$\Rightarrow r = \sqrt{a^2+b^2} = (a^2+b^2)^{1/2}$$

& $\theta = \tan^{-1}(b/a)$

(1) $\Rightarrow y_1 = e^{ax} [r \sin(bx+c) \cdot \cos\theta + b \cdot \cos(bx+c) \cdot \sin\theta]$

$$= e^{ax} r [\sin(bx+c+\theta)]$$

$\because \sin(A+B) = \sin A \cos B + \cos A \sin B$

again diff. w.r.to x. we get

$$y_2 = r^2 e^{ax} \sin(bx+c+2\theta)$$

$$y_3 = r^3 e^{ax} \sin(bx+c+3\theta)$$

$$\vdots$$

$$y_n = r^n \cdot e^{ax} \cdot \sin(bx+c+n\theta)$$

i.e. $$\boxed{y_n = (a^2+b^2)^{n/2} \cdot e^{ax} \sin[bx+c+n\tan^{-1}(b/a)]}$$

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(8) let $y = e^{ax} \cdot \cos(bx+c)$

$\Rightarrow y_n = (a^2+b^2)^{n/2} e^{ax} \cos\left[bx+c+n \cdot \tan^{-1}(b/a)\right]$

---

(9) If $y = \log(ax+b)$

Diff w.r. to x we get

$$y_1 = \frac{1}{(ax+b)} \cdot a$$

again diff. w.r.to x, $(n-1)$ time

$$y_n = a \cdot \frac{d^{n-1}}{dx^{n-1}} \left[\frac{1}{(ax+b)}\right]$$

$$= a \cdot \left[\frac{(-1)^{n-1} \cdot (n-1)! \; a^{n-1}}{(ax+b)^n}\right]$$

$$\boxed{y_n = \frac{(-1)^{n-1} \cdot (n-1)! \; a^n}{(ax+b)^n}}$$

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Ex: find n^{th} derivative of following function

(i) $y = \cos^4 x$

(ii) $y = \sin 2x \cdot \sin 3x$

(iii) $y = \dfrac{x^2+4}{(2x+3)\cdot(x-1)^2}$

Soln (i). let $y = \cos^4 x$

$$= \left[\frac{2\cos^2 x}{2}\right]^2 = \left[\cos^2 x\right]^2$$

$$= \left[\frac{1+\cos 2x}{2}\right]^2$$

$$= \frac{1}{4}(1+\cos 2x)^2$$

$$= \frac{1}{4}\left[1 + 2\cos(2x) + \cos^2(2x)\right]$$

$$= \frac{1}{4}\left[1 + 2\cos(2x) + \left(\frac{1+\cos 4x}{2}\right)\right]$$

$$\therefore y = \frac{1}{8}\left[3 + \cos(4x) + 4\cos(2x)\right]$$

Differentiating n^{th} time we get

$$y_n = \frac{1}{8}\frac{d^n}{dx^n}\left[3 + \cos(4x) + 4\cos(2x)\right]$$

$$= \frac{1}{8}\left[0 + \frac{d^n}{dx^n}(\cos 4x) + 4\frac{d^n}{dx^n}(\cos 2x)\right]$$

$$\therefore y_n = \frac{1}{8}\left[4^n \cos\left(4x + n\pi/2\right) + 4\cdot(2^n)\cos\left(2x + \frac{n\pi}{2}\right)\right]$$

$$\because \quad D^n[\cos(ax+b)] = a^n \cos\left(ax+b+n\pi/2\right)$$

Soln (ii) let $y = \sin 2x \cdot \sin 3x$

$$= \frac{1}{2}\left[\cos(2x-3x) - \cos(2x+3x)\right]$$

$$= \frac{1}{2}\left[\cos(-x) - \cos 5x\right]$$

$$= \frac{1}{2}\left[\cos x - \cos 5x\right]$$

Diff. n^{th} time w.r.to x. we get

$$y_n = \frac{1}{2}\left[D^n(\cos x) - D^n(\cos 5x)\right]$$

$$= \frac{1}{2}\left[(1)^n \cos(x + n\pi/2) - 5^n \cos(5x + n\pi/2)\right]$$

$$\therefore y_n = \frac{1}{2}\left[\cos(x + n\pi/2) - 5^n \cos(5x + n\pi/2)\right]$$

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Soln ③ let $y = \frac{x^2+4}{(2x+3)(x-1)^2}$

or $\frac{A}{(2x+3)} + \frac{Bx+C}{(x-1)^2}$

$$= \frac{A}{(2x+3)} + \frac{B}{(x-1)} + \frac{C}{(x-1)^2} \quad \text{—— (*)}$$

$$\Rightarrow x^2+4 = A(x-1)^2 + B(x-1)(2x+3) + C(2x+3)$$

$$= (A+2B)x^2 + (-2A+B+2C)x + (A-3B+3C)$$

Equating like Term of like power of x on b.s. we get

$$A+2B = 1 \Rightarrow A = 1-2B \quad \text{—— (1)}$$
$$-2A+B+2C = 0 \quad \text{—— (2)}$$
$$A-3B+3C = 4 \quad \text{—— (3)}$$

Solving these equations we get

$$A = 1, \quad B = 0, \quad C = 1$$

$$\therefore y = \frac{1}{2x+3} + \frac{1}{(x-1)^2}$$

Differentiating n^{th} time, w.r.to x, we get

$$y_n = D^n\left[\frac{1}{2x+3}\right] + D^n\left[\frac{1}{(x-1)^2}\right]$$

$$= \frac{(-1)^n\, n!\, 2^n}{(2x+3)^{n+1}} + \frac{(-1)^n\,(n+1)!\,(1)^n}{(2-1)!\,(x-1)^{n+2}}$$

$$\Rightarrow y_n = (-1)^n\left[\frac{n!\,2^n}{(2x+3)^{n+1}} + \frac{(n+1)!}{(x-1)^{n+2}}\right]$$

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Ex: If $y = \tan^{-1}\left(\frac{x}{a}\right)$

Prove that

$$y_n = \frac{(-1)^{n-1}\,(n-1)!}{a^n} \cdot \sin(n\theta) \cdot \sin^n\theta$$

Soln: let $y = \tan^{-1}(x/a)$ — (1)

Diff. w.r.to. x. we get

$$y_1 = \frac{d}{dx}\tan^{-1}(x/a) = \frac{1}{1+(x/a)^2} \cdot \left(\frac{1}{a}\right)$$

$$\Rightarrow y_1 = \frac{a}{x^2+a^2}$$

$$\Rightarrow y_1 = \frac{a}{(x+ia)(x-ia)}$$

$$= \frac{1}{2i}\left[\frac{1}{(x-ia)} - \frac{1}{(x+ia)}\right] \quad \text{— (2)}$$

Diff. eqn (2) (n-1) times w.r to. x. we get

$$y_n = \frac{1}{2i}\left\{D^{n-1}\left[\frac{1}{x-ia}\right] - D^{n-1}\left[\frac{1}{x+ia}\right]\right\}$$

$$= \frac{1}{2i}\left[\frac{(-1)^{n-1}\cdot(n-1)!}{(x-ia)^n} - \frac{(-1)^{n-1}\cdot(n-1)!}{(x+ia)^n}\right]$$

OR Show that $y_n = \frac{(-1)^{n-1}(n-1)!}{x^n} \cdot \sin n\theta \cdot \cos^n\theta$

OR Show that $y_n = (-1)^{n-1}(n-1)! \sin n\theta \ (q^2+x^2)^{-n/2}$

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$$\therefore y_n = \frac{1}{2i}\left[\frac{(-1)^n(n-1)!}{2i}\right]\left(\frac{1}{(x-iq)^n} - \frac{1}{(x+iq)^n}\right)$$

put $x = r\cos\theta$, $q = r\sin\theta$

$\therefore r = \sqrt{x^2+q^2}$, & $\theta = \tan^{-1}(x/q)$

also $x-iq = r\cos\theta - ir\sin\theta = r(\cos\theta - i\sin\theta) = re^{-i\theta}$

$$\therefore y_n = \frac{(-1)^n(n-1)!}{2i}\left[\frac{1}{(r\cos\theta - ir\sin\theta)^n} - \frac{1}{(r\cos\theta + ir\sin\theta)^n}\right]$$

$$= \frac{(-1)^n(n-1)!}{2i}\frac{1}{r^n}\left[\frac{1}{(\cos\theta - i\sin\theta)^n} - \frac{1}{(\cos\theta + i\sin\theta)^n}\right]$$

$$= \frac{(-1)^n(n-1)!}{2i}\frac{1}{r^n}\left[\frac{1}{\cos n\theta - i\sin n\theta} - \frac{1}{\cos n\theta + i\sin n\theta}\right]$$

(∵ By De Moivre's Thm)

$$= \frac{(-1)^n(n-1)!}{2i}\frac{1}{r^n}\left[\frac{2i\sin n\theta}{\cos^2 n\theta + \sin^2 n\theta}\right]$$

$$= \frac{(-1)^n(n-1)!}{r^n}\cdot\sin n\theta$$

$$= (-1)^n(n-1)!\ \sin(n\theta)\cdot r^{-n}$$

$$= (-1)^n(n-1)!\ \sin n\theta \cdot \left(\frac{\sin\theta}{q}\right)^n \qquad \because q = r\sin\theta \Rightarrow r = \frac{q}{\sin\theta}$$

$$\therefore \boxed{y_n = \frac{(-1)^n(n-1)!\sin n\theta \cdot \sin^n\theta}{q^n}}$$ H.P.

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☆ Leibnitz's Theorem

If $y = u \cdot v$, where $u = f(x)$, & $v = g(x)$ are any two functions of x. Then

$$D^n y = y_n = (u \cdot v)_n$$

$$\Rightarrow \boxed{(u \cdot v)_n = u v_n + n \cdot u_1 v_{n-1} + \frac{n(n-1)}{2!} u_2 v_{n-2} + \frac{n(n-1)(n-2)}{3!} u_3 \cdot v_{n-3} + \cdots\cdots + u_n \cdot v.}$$

For eg: If $y = (1+x)^2 \cdot e^{ax}$, find y_n

Soln: let $u = (1+x)^2$, $v = e^{ax}$

$\Rightarrow u_1 = 2(1+x)$, $v_1 = a e^{ax}$

$u_2 = 2$, $v_2 = a^2 e^{ax}$

$u_3 = 0$, $v_3 = a^3 e^{ax}$

$\vdots$, $\vdots$

$u_n = 0$, $v_n = a^n e^{ax}$

Now.

$$y = u \cdot v = (1+x)^2 \cdot e^{ax} \quad \text{——— (1)}$$

Diff. eqn (1) w.r.to x. n^{th} time using Leibnitz's thm

$$y_n = uv_n + n u_1 v_{n-1} + \frac{n(n-1)}{2!} u_2 v_{n-2} + \cdots + u_n \cdot v$$

$$= (1+x)^2 a^n e^{ax} + n \cdot 2(1+x) a^{n-1} e^{ax} + \frac{n(n-1)}{2!} \cdot 2 \cdot a^{n-2} e^{ax} + 0 + \cdots + 0$$

$$\therefore y_n = (1+x)^2 a^n e^{ax} + 2n(1+x) a^{n-1} e^{ax} + n(n-1) a^{n-2} e^{ax}$$

$$= a^n e^{ax} \left[(1+x)^2 + \frac{2n(1+x)}{a} + \frac{n(n-1)}{a^2} \right]$$

Ex (ii) let $y = (1+3x^2) \cdot \sin 2x$ find y_n

let $u = (1+3x^2)$ $\quad v = \sin 2x$

$\Rightarrow u_1 = 6x$ $\quad v_1 = 2\cos 2x = 2\sin(2x+\pi/2)$

$u_2 = 18x$ $\quad v_2 = 4\cos(2x+\pi/2) = 4\sin(2x+\pi)$

$u_3 = 18$ $\quad = 2^2 \sin(2x+2\pi/2)$

$u_4 = 0$ $\quad \vdots$

$u_5 = 0$ $\quad v_{n-1} = 2^{n-1} \cdot \sin(2x + (n-1)\pi/2)$

$\vdots$ $\quad v_n = 2^n \sin(2x + n\pi/2)$

$u_n = 0$

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Now using Leibnitz thm

$$y_n = (u \cdot v)_n$$

$$= u \cdot v_n + n\, u_1 v_{n-1} + \frac{n(n-1)}{2!} u_2 v_{n-2}$$

$$+ \frac{n(n-1)(n-2)}{3!} u_3 v_{n-3} + \frac{n(n-1)(n-2)(n-3)}{4!} u_4 v_{n-4}$$

$$+ \ldots + u_n v$$

$$= (1+3x^3)\, 2^n \sin(2x + n\pi/2)$$

$$+ n \cdot (9x^2)\, 2^{n-1} \sin(2x + (n-1)\pi/2)$$

$$+ \frac{n(n-1)}{2!} (18x)\, 2^{n-2} \sin(2x + (n-2)\pi/2)$$

$$+ \frac{n(n-1)(n-2)}{3!} (18) \cdot 2^{n-3} \sin(2x + (n-3)\pi/2)$$

$$+ 0 + \ldots + 0$$

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Ex: If $y = \cos(m \sin^{-1} x)$, prove that

$$(1-x^2) y_{n+2} - (2n+1) x \cdot y_{n+1} + (m^2 - n^2) y_n = 0$$

Soln: let $y = \cos(m \sin^{-1} x)$ —— (1)

Diff. w.r to x. we get

$$y_1 = -\sin(m \cdot \sin^{-1} x) \cdot m \cdot \frac{1}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = -m \cdot \sin(m \sin^{-1} x)$$

squaring on b.s. we get

$$(1-x^2) y_1^2 = m^2 \sin^2(m \sin^{-1} x)$$

$$= m^2 [1 - \cos^2(m \sin^{-1} x)]$$

$$\Rightarrow (1-x^2) y_1^2 = m^2 [1 - y^2] \quad \because \text{By eqn (1)}$$

—— (2)

again diff. w.r to x we get

$$(1-x^2) 2y_1 \cdot y_2 + (-2x) y_1^2 = m^2 [0 - 2y \cdot y_1]$$

$$\Rightarrow 2y_1 [(1-x^2) y_2 - x y_1] = -2y_1 \cdot (m^2 y)$$

$$\Rightarrow (1-x^2) y_2 - x y_1 + m^2 y = 0 \quad \text{—— (3)}$$

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Using Leibnitz thm. we get (Diff. n^{th} time w.r.to.x)

$$D^n[(1-x^2)y_2] - D^n[xy_n] + m^2 D^n[y] = 0$$

$$\Rightarrow \left\{(1-x^2)y_{n+2} + n\cdot(-2x)y_{n+1} + \frac{n(n-1)}{2!}(-2)y_n\right\} - [xy_{n+1} + n\cdot(1)y_n] + m^2 y_n = 0$$

$$\Rightarrow \boxed{(1-x^2)y_{n+2} - (2n+1)\cdot x\, y_{n+1} + (m^2-n^2)y_n = 0} \quad \text{H.P.}$$

Ex: If $y = a\cos(\log x) + b\sin(\log x)$

Show that (i) $x^2y_2 + xy_1 + y = 0$

(ii) $x^2y_{n+2} + (2n+1)xy_{n+1} + (n^2+1)y_n = 0$

Soln:

Let $y = a\cos(\log x) + b\sin(\log x)$ ——(1)

Diff. w.r.to x. we get

$$y_1 = a\frac{[-\sin(\log x)]}{x} + b\frac{\cos(\log x)}{x}$$

$$\Rightarrow xy_1 = -a\sin(\log x) + b\cos(\log x)$$

again diff. w.r.to x. we get

$$\Rightarrow x y_2 + y_1 = -a\frac{\cos(\log x)}{x} - b\frac{\sin(\log x)}{x}$$

$$\Rightarrow x^2 y_2 + x y_1 = -[a\cos(\log x) + b\sin(\log x)]$$

$$\Rightarrow x^2 y_2 + x y_1 = -y \qquad (\because \text{using } ①)$$

$$\Rightarrow \boxed{x^2 y_2 + x y_1 + y = 0} \qquad \text{—②}$$

Diff. n^{th} time w.r.to. x using leibnitz thm.

$$D^n[x^2 y_2] + D^n[x y_1] + D^n[y] = 0$$

$$\Rightarrow \left[x^2 y_{n+2} + n\cdot(2x)\, y_{n+1} + \frac{n(n-1)}{2!}(2)\cdot y_n\right]$$

$$+ \left[x y_{n+1} + n(1) y_n\right] + y_n = 0$$

$$\Rightarrow \boxed{x^2 y_{n+2} + (2n+1)x y_{n+1} + (n^2+1) y_n = 0}$$

H.P.

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Ex: If u_n denotes the n^{th} derivative of $\frac{Lx+M}{x^2-2Bx+C}$ prove that

$$\frac{x^2-2Bx+C}{(n+1)(n+2)}u_{n+2} + \frac{2(x-B)}{n+1}u_{n+1} + u_n = 0$$

Soln $\Rightarrow$ Let $u = \frac{Lx+M}{x^2-2Bx+C}$ —— (1)

$\Rightarrow (x^2-2Bx+C)u = Lx+M$

Diff. w.r to x. we get

$(x^2-2Bx+C)u_1 + 2(x-B)u = L$ —— (2)

again diff. w.r to x. we get

$(x^2-2Bx+C)u_2 + 2(x-B)u_1 + 2(x-B)u_1 + 2u = 0$

$\Rightarrow (x^2-2Bx+C)u_2 + 4(x-B)u_1 + 2u = 0$ —— (3)

Diff. n^{th} time. w.r.to x, using Leibnitz th^m.

$D^n[(x^2-2Bx+C)\cdot u_2] + 4D^n[(x-B)u_1] + 2D^n[u] = 0$

$\Rightarrow \{(x^2-2Bx+C)u_{n+2} + n(2x-2B)\cdot u_{n+1} + \frac{n(n-1)}{2!}(2)\cdot u_n\}$

$+ 4\{(x-B)u_{n+1} + n(1)\cdot u_n\} + 2u_n = 0$

$$\Rightarrow (x^2-2Bx+C)\,u_{n+2} + n(2n+4)(x-B)\,u_{n+1} + (n^2+3n+2)\,u_n = 0$$

Dividing b.s. by (n^2+3n+2)

$$\frac{(x^2-2Bx+C)}{(n^2+3n+2)}u_{n+2} + \frac{2(n+2)(x-B)}{(n^2+3n+2)}u_{n+1} + u_n = 0$$

$$\Rightarrow \boxed{\frac{x^2-2Bx+C}{(n+1)(n+2)}u_{n+2} + \frac{2(x-B)}{(n+1)}u_{n+1} + u_n = 0}$$

H.P.

Ex: If $y^{1/m} + y^{-1/m} = 2x$

prove that (i) $(x^2-1)y_2 + xy_1 - m^2 y = 0$

(ii) $(x^2-1)y_{n+2} + (2n+1)x\,y_{n+1} + (n^2-m^2)y_n = 0$

Soln: let $y^{1/m} + y^{-1/m} = 2x$ —(1)

$\Rightarrow$ let $y^{1/m} = z \Rightarrow y^{-1/m} = 1/z$

$\therefore$ (1) $\Rightarrow z + \frac{1}{z} = 2x$

$\Rightarrow z^2 - 2xz + 1 = 0$

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$\therefore \quad z = \dfrac{+(2x) \pm \sqrt{(2x)^2 - 4(1)(1)}}{2(1)}$

$= \dfrac{+2x \pm \sqrt{4x^2-4}}{2} \qquad = -x \pm \sqrt{x^2-1}$

$\therefore \quad y^{1/m} = +x \pm \sqrt{x^2-1}$

ie. $y = \left(+x \pm \sqrt{x^2-1}\right)^m$ ——— (1)

Diff. eqn (1) w.r.to x. we get

$$y_1 = m\left(+x \pm \sqrt{x^2-1}\right)^{m-1} \cdot \left[+1 \pm \frac{1 \cdot 2x}{2\sqrt{x^2-1}}\right]$$

$$= m\left[+x \pm \sqrt{x^2-1}\right]^{m-1} \cdot \frac{\left(+\sqrt{x^2-1} \pm x\right)}{\sqrt{x^2-1}}$$

$$\Rightarrow \sqrt{x^2-1} \cdot y_1 = m\left(+x \pm \sqrt{x^2-1}\right)^{m-1} \cdot \left[\pm\left(+x \pm \sqrt{x^2-1}\right)\right] \quad (\pm 1)$$

$$\Rightarrow \sqrt{x^2-1} \cdot y_1 = m\,(\pm 1)\left(x \pm \sqrt{x^2-1}\right)^m$$

$\Rightarrow \sqrt{x^2-1} \cdot y_1 = m \cdot y$

$\Rightarrow$ squaring on b.s. we get

$(x^2-1)\,y_1^2 = m^2 y^2$

$\Rightarrow$ Diff. w.r.to x. we get

$(x^2-1)\,2y_1 \cdot y_2 + (2x) \cdot y_1^2 = m^2 \cdot 2y \cdot y_1$

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$\Rightarrow (x^2-1)\, y_2 + x\, y_1 = m^2 y$

$\Rightarrow (x^2-1)\, y_2 + x\, y_1 - m^2 y = 0$ —(2)

Diff n^{th} times w.r.to. x using Leibnitz rule

$$D^n[(x^2-1)\, y_2] + D^n[x\, y_1] - m^2\, D^n[y] = 0$$

$$\Rightarrow \left\{(x^2-1)\, y_{n+2} + n\,(2x)\, y_{n+1} + \frac{n(n-1)}{2!}(2)\, y_n\right\}$$

$$+ \left\{x\, y_{n+1} + n(1)\, y_n\right\} - m^2\, y_n = 0$$

$$\Rightarrow \boxed{(x^2-1)\, y_{n+2} + (2n+1)\, x\, y_{n+1} + (n^2-m^2)\, y_n = 0}$$

H.P.

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Ex If $y = e^{m\cos^{-1}x}$,

prove that $(1-x^2)\,y_{n+2} - (2n+1)\,x\,y_{n+1} - (m^2+n^2)\,y_n = 0$

Hence find y_n at $x=0$ i.e. $y_n(0)$.

Soln: Let $y = e^{m\cos^{-1}x}$ ——— (1)

Diff. w.r.to x. we get

$$y_1 = e^{m\cos^{-1}x} \cdot m\cdot\left(\frac{-1}{\sqrt{1-x^2}}\right).$$

$$\Rightarrow \sqrt{1-x^2}\,y_1 = -m\,y. \quad \text{——— (2)}$$

Squaring on both side we get

$$(1-x^2)\,y_1^2 = m^2 y^2 \quad \text{——— (2)}$$

again diff. w.r to x., we get

$$(1-x^2)2y_1\cdot y_2 + (-2x)\,y_1^2 = 2m^2\,y\cdot y_1$$

$$\Rightarrow (1-x^2)\,y_2 - x\,y_1 = m^2 y$$

$$\Rightarrow (1-x^2)\,y_2 - x\,y_1 - m^2 y = 0 \quad \text{——— (3)}$$

Diff. n^{th} times w.r.to x using Leibnitz th^m

$$D^n\left[(1-x^2)y_2\right] - D^n\left[x y_n\right] - m^2 D^n\left[y_n\right] = 0$$

$$\Rightarrow \left\{(1-x^2)\,y_{n+2} + n(-2x)\,y_{n+1} + \frac{n(n-1)}{2!}(-2)\,y_n\right\} - \left\{x\,y_{n+1} + n(1)\cdot y_n\right\} - m^2 y_n = 0$$

$$\Rightarrow \boxed{(1-x^2)\,y_{n+2} - (2n+1)\,x\,y_{n+1} - (n^2+m^2)\,y_n = 0} \quad \text{---(4)}$$

H.P.

To find $y_n(0)$ i.e. y_n at $x=0$

put $x=0$ in eqn (1), (2), (3) & (4) we get

(1) $\Rightarrow$ $y(0) = e^{m\cos^{-1}(0)} = e^{m\pi/2}$

(2) $\Rightarrow$ $y_1(0) = -m\,y(0) = -m\,e^{m\pi/2}$

(3) $\Rightarrow$ $y_2(0) = m^2\,y(0) = m^2\,e^{m\pi/2}$

& (4) $\Rightarrow$ $\boxed{y_{n+2}(0) = (n^2+m^2)\,y_n(0)}$ ——— (5)

Now. If n is odd i.e. $n = 1, 3, 5, \ldots$

For $n=1$ $y_3(0) = (1^2+m^2)\,y_1(0) = (1^2+m^2)(-m)\,e^{m\pi/2}$

For $n=3$ $y_5(0) = (3^2+m^2)\,y_3(0)$

$= (3^2+m^2)(1^2+m^2)(-m)\cdot e^{m\pi/2}$

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n = 5.

$$y_7(0) = (5^2+m^2)\cdot y_5(0)$$

$$= (5^2+m^2)(3^2+m^2)(1^2+m^2)\cdot(-m)\cdot e^{m\pi/2}$$

$$\vdots$$

$$y_n(0) = [(n-2)^2+m^2]\ldots(5^2+m^2)(1^2+m^2)(-m)\,e^{m\pi/2}$$

If n is even : ie. n = 2, 4, 6, ...

for n = 2 :

$$y_4(0) = (2^2+m^2)\,y_2(0)$$

$$= (2^2+m^2)\,m^2\,e^{m\pi/2}$$

For n = 4

$$y_6(0) = (4^2+m^2)\cdot y_4(0)$$

$$= (4^2+m^2)(2^2+m^2)\cdot m^2\,e^{m\pi/2}$$

$$\vdots$$

$$y_n(0) = ((n-2)^2+m^2)\ldots(2^2+m^2)\,m^2\,e^{m\pi/2}$$

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H.W

(1) If $y = \sin \log (x^2+2x+1)$, prove that

(i) $(x+1)^2 y_2 + (x+1) y_1 + 4y = 0$

(ii) $(x+1)^2 y_{n+2} + (2n+1)(x+1) y_{n+1} + (n^2+4) y_n = 0$

(2) If $y = \tan^{-1} x$, prove that.

$(1+x^2) y_{n+2} + 2(n+1) x y_{n+1} + n(n+1) y_n = 0$

(3) If $\cos^{-1}\left(\frac{y}{b}\right) = \log (x/n)^n$

Show that $x^2 y_{n+2} + (2n+1) x \cdot y_{n+1} + 2n^2 y_n = 0$

(4) If $y = \left(x + \sqrt{x^2+a^2}\right)^m$

Show that

$(x^2+a^2) y_{n+2} + (2n+1) x y_{n+1} + (n^2-m^2) y_n = 0$

Hence find y_n at $x=0$

(5) If $y = [\sin^{-1} x]^2$, prove that

$(1-x^2) y_{n+2} - (2n+1) x y_{n+1} - n^2 y_n = 0$

Hence find $y_n(0)$.

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★ TAYLOR'S AND MACLAURINS SERIES FOR ONE VARIABLE

Taylor's Thm

Statement :

If $f(x+h)$ can be expanded in a series of ascending power of x, then the expansion is

$$f(x+h) = f(h) + x f'(h) + \frac{x^2}{2!} f''(h) + \frac{x^3}{3!} f'''(h) + \dots\dots$$

Case ① : Interchanging x and h, we get

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots\dots$$

This is the Taylor's series in ascending power of h.

Case (ii) : Replacing x by a and h by $(x-a)$, we get

$$f(x) = f(a) + (x-a) f'(a) + \frac{(x-a)^2}{2!} f''(a) + \frac{(x-a)^3}{3!} f'''(a) + \dots$$

This represent expansion of $f(x)$ in power of $(x-a)$ [i.e. about pt a.

Case (iii) : If $a=0$, then

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots\dots$$

This represent expansion of $f(x)$ in ascending power of x. about origin

known as Maclaurin's expansion of $f(x)$

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Some standard expansions (using Maclaurin's series)

(1) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + \frac{x^5}{5!} + \dots$

(2) $e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \frac{x^4}{4!} - \frac{x^5}{5!} + \dots$

(3) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} - \dots$

(4) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \dots$

(5) $\sinh x = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{x^7}{7!} + \dots$

(6) $\cosh x = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \dots$

(7) $\tan x = x + \frac{x^3}{3} + \frac{2}{15}x^5 + \dots$

(8) $\tanh x = x - \frac{x^3}{3} + \frac{2}{15}x^5 - \dots$

(9) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \frac{x^5}{5} - \dots$

(10) $a^x = 1 + x(\log a) + \frac{x^2}{2!}(\log a)^2 + \frac{x^3}{3!}(\log a)^3 + \dots$

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(11) $(1+x)^m = 1 + m \cdot x + \frac{m(m-1)}{2!} x^2 + \frac{m(m-1)(m-2)}{3!} x^3 + \ldots$

(12) $(1+x)^{-1} = 1 - x + x^2 - x^3 + x^4 - x^5 + \ldots$

(13) $(1+x)^{-1} = 1 + x + x^2 + x^3 + x^4 + \ldots$

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Taylors series

Ex: Expand $\cos x$ in power of $(x-\pi/4)$ up to 4 terms

Soln: Let $f(x) = \cos x$

By using taylors series expansion of $f(x)$ in power of $(x-a)$ we have

$$f(x) = f(a) + (x-a)f'(a) + \frac{(x-a)^2}{2!}f''(a) + \frac{(x-a)^3}{3!}f'''(a) + \dots \quad \text{---(1)}$$

Put $a = \pi/4$

Since

$f(x) = \cos x \Rightarrow f(\pi/4) = \cos \pi/4 = 1/\sqrt{2}$

$f'(x) = -\sin x \Rightarrow f'(\pi/4) = -1/\sqrt{2}$

$f''(x) = -\cos x \Rightarrow f''(\pi/4) = -1/\sqrt{2}$

$f'''(x) = \sin x \Rightarrow f'''(\pi/4) = 1/\sqrt{2}$

putting these values in (1)

$$f(x) = \cos x = f(\pi/4) + (x-\pi/4)f'(\pi/4) + \frac{(x-\pi/4)^2}{2!}f''(\pi/4) + \frac{(x-\pi/4)^3}{3!}f'''(\pi/4) + \dots$$

$$= 1/\sqrt{2} + (x-\pi/4)(-1/\sqrt{2}) + \frac{(x-\pi/4)^2}{2!}(-1/\sqrt{2}) + \frac{(x-\pi/4)^3}{3!}(1/\sqrt{2}) + \dots$$

$$\cos x = \frac{1}{\sqrt{2}}\left[1 - (x-\pi/4) - \frac{1}{2!}(x-\pi/4)^2 + \frac{1}{3!}(x-\pi/4)^3 + \dots\right]$$

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Ex: Given $f(x) = x^3 + 8x^2 + 15x - 24$

find $f(\frac{11}{10})$ by using Taylors series.

Soln: We know that.

Taylor series expansion of $f(x+h)$ in power of h is

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \cdots \quad \text{--- (1)}$$

let us consider

$$x + h = 11/10$$

$$= 1 + \frac{1}{10} = 1 + 0.1$$

Cosider $x = 1$ and $h = 0.1$

Thus

$$(1) \Rightarrow f\left(\frac{11}{10}\right) = f(1) + (0.1) f'(1) + \frac{(0.1)^2}{2!} f''(1) + \frac{(0.1)^3}{3!} f'''(1) + \cdots \quad \text{--- (2)}$$

Now

$$f(x) = x^3 + 8x^2 + 15x - 24 \Rightarrow f(1) = 0$$

$$f'(x) = 3x^2 + 16x + 15 \Rightarrow f'(1) = 34$$

$$f''(x) = 6x + 16 \Rightarrow f''(1) = 22$$

$$f'''(x) = 6 \Rightarrow f'''(1) = 6$$

$$(2) \Rightarrow f\left(\frac{11}{10}\right) = 0 + (0.1)(34) + \frac{(0.1)^2}{2!}(22) + \frac{(0.1)^3}{3!}(6) + 0$$

$$\boxed{f\left(\frac{11}{10}\right) = 3.511}$$

Ex Expand $\tan(x + \pi/4)$ up to the term of x^4 and evaluate $\tan(46^\circ)$ to four significant digits.

Soln Let

$$f(x+h) = \tan(x+h) = \tan(x + \pi/4)$$

Here $h = \pi/4$

Thus

$$f(x) = \tan(x) \Rightarrow f(\pi/4) = \tan \pi/4 = 1$$

$$f'(x) = \sec^2 x = 1 + \tan^2 x \Rightarrow f'(\pi/4) = 2$$

$$f''(x) = 2 \tan x \sec^2 x$$
$$= 2\tan x + 2\tan^3 x \Rightarrow f''(\pi/4) = 4$$

$$f'''(x) = 2\sec^2 x + 6 \tan x \sec^2 x$$
$$= 2(1 + \tan^2 x) + 6\tan x (1 + \tan^2 x)$$

$$f^{IV}(x) = 16 \tan x + 16\tan^3 x + 2 + 6\tan^2 x \Rightarrow f'''(\pi/4) = 16$$
and so on. $\Rightarrow f^{IV}(\pi/4) = 80$

Now By Taylor series expansion.

$$f(x+h) = f(h) + x f'(h) + \frac{x^2}{2!} f''(h) + \frac{x^3}{3!} f'''(h) + \frac{x^4}{4!} f^{IV}(h) + \dots$$

$$\Rightarrow \tan(x + \pi/4) = f(\pi/4) + x \cdot f'(\pi/4) + \frac{x^2}{2!} f''(\pi/4) + \frac{x^3}{3!} f'''(\pi/4)$$
$$+ x^4/4! \, f^{IV}(\pi/4) + \dots$$

$$\tan(x + \pi/4) = 1 + x(2) + \frac{x^2}{2!}(4) + \frac{x^3}{3!}(16) + \frac{x^4}{4!}(80) + \dots$$

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How to find $\tan(46.5^\circ)$

let $x + \pi/4 = 46.5^\circ$

$= 45^\circ + 1.5^\circ$

$= \pi/4 + 1.5^\circ$

$\Rightarrow x = 1.5^\circ = 1.5 \times \frac{\pi}{180} = 0.02618$ radian

put $x = 0.02618$ in (1) we get

$\tan(46.5^\circ) = \tan(\pi/4 + (1.5^\circ))$

$= \tan(\pi/4 + 0.02618)$

$= 1 + (0.02618)(2) + \frac{(0.02618)^2}{2!}(4)$

$+ \frac{(0.02618)^3}{6}(16) + \frac{(0.02618)^4}{24}(80) + \cdots$

$= 1.05377786$

≈ 1.0538

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H.W :

① Using taylor series expansion evaluate

Sin 60°.30' correct to five decimal places.

Given $\sin 60° = 0.86603$

$1° = 0.01745$ Ans 0.87036

Hint $x+h = 60°.30' = (60° + \frac{1}{2}°)$

$= (x+h)$

$\therefore x = 60° = \pi/3$

$h = \frac{1}{2}° = \left(\frac{0.01745}{2}\right)$ radian.

② Calculate approximate value of $\sqrt{10}$ to four decimal places by taking the first four term of an appropriate Taylor's series expansion.

Hint ~~$f(x) = \sqrt{10} = x \Rightarrow f(x) = x^{1/2} = 10$~~

$\therefore$ ~~$f(x+h) = (x+h)^{1/2}$~~

let $f(x) = x^{1/2}$ (sq. root of x)

$\Rightarrow f(x+h) = (x+h)^{1/2}$

put

$x+h = 10 = 9+1$

$\Rightarrow x = 9, \; h = 1$ Ans 3.1623

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(3) Using Taylor series expansion find the value of ~~Cos 45°~~ Cos(64°) correct to four decimal places.

Ans · 0.4383

Hint $x+h = 64°$

$= 60 + 4°$

$= \pi/3 + \frac{4\pi}{180}$

$= \pi/3 + 0.069813$

$\Rightarrow$ let $x = \pi/3$ & $h = 0.069813$.

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Maclaurin's series

Ex: prove that

$$\log(\sec x + \tan x) = x + \frac{x^3}{6} + \frac{x^5}{24} + \cdots$$

Soln: ⇒ Let $f(x) = \log(\sec x + \tan x)$

By Maclaurin's series expansion.

$$f(x) = f(0) + x f'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \frac{x^4}{4!} f^{iv}(0) + \frac{x^5}{5!} f^{v}(0) + \cdots \quad \text{—①}$$

Here.

$f(x) = \log(\sec x + \tan x) \Rightarrow f(0) = \log(\sec 0 + \tan 0) = 0$

$f'(x) = \dfrac{\sec x \cdot \tan x + \sec^2 x}{(\sec x + \tan x)}$

$= \sec x \Rightarrow f'(0) = \sec(0) = 1$

$f''(x) = \sec x \cdot \tan x \Rightarrow f''(0) = \sec(0) \cdot \tan(0) = 0$

$f'''(x) = \sec x \cdot \sec x^2 + \tan x \cdot (\sec x \cdot \tan x)$

$= \sec x \cdot [\sec^2 x + \tan^2 x]$

$= \sec x \cdot [\sec^2 x + \sec^2 x - 1] \Rightarrow f'''(0) = 1$

$= 2\sec^3 x - \sec x.$

$f^{iv}(x) = 6\sec^2 x \cdot \sec x \cdot \tan x - \sec x \cdot \tan x$

$= [6\sec^2 x - 1] \sec x \cdot \tan x \Rightarrow f^{iv}(0) = 0$

$f^{v}(x) = (6\sec^2 x - 1)\cdot(\sec x \cdot \sec^2 x + \tan x \cdot \sec x \cdot \tan x)$

$+[(12\sec x)\cdot \sec x \cdot \tan x]\ \sec x \cdot \tan x$

$\Rightarrow f^{v}(0) = 5.$

Putting these values in eqn (1), we get

$$f(x) = 0 + x + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(1) + \frac{x^4}{4!}(0) + \frac{x^5}{5!}(5) + \dots$$

$$= x + \frac{x^3}{6} + \frac{x^5}{24} + \dots$$

Thus. $\log(\sec x + \tan x) = x + x^3/3 + x^5/24 + \dots$

Ex ⇒ prove that

$$\log(1+e^x) = \log 2 + \frac{x}{2} + \frac{x^2}{8} - \frac{x^4}{192} - \dots$$

OR

Expand $\log(1+e^x)$ in ascending powers of x upto term of x^4.

Soln

let $f(x) = \log(1+e^x)$

By Maclaurin's series expansion

$$\log(1+e^x) = f(x) = f(0) + x f'(x) + \frac{x^2}{2!} f''(x) + \frac{x^3}{3!} f'''(x) + \frac{x^4}{4!} f^{iv}(x) + \dots$$

(1)

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Here

$$f(x) = \log(1+e^x) \Rightarrow f(0) = \log(1+e^0) = \log(1+1) = \log 2$$

$$f'(x) = \frac{1}{1+e^x} e^x \Rightarrow f'(0) = \frac{e^0}{1+e^0} = \frac{1}{2}$$

$$f''(x) = \frac{(1+e^x)\cdot e^x - e^x(e^x)}{(1+e^x)^2}$$

$$= \frac{e^x}{(1+e^x)^2} \Rightarrow f''(0) = \frac{1}{(1+1)^2} = \frac{1}{4}$$

$$f'''(x) = \frac{(1+e^x)^2\cdot e^x - 2(1+e^x)\cdot e^x \cdot e^x}{(1+e^x)^4}$$

$$= \frac{(1+e^x)\left[(1+e^x)e^x - 2e^{2x}\right]}{(1+e^x)^4}$$

$$= \frac{e^x + e^{2x} - 2e^{2x}}{(1+e^x)^3}$$

$$= \frac{e^x - e^{2x}}{(1+e^x)^3} \Rightarrow f'''(0) = \frac{e^0 - e^0}{(1+e^0)^3} = \frac{0}{8}$$

$$= 0$$

$$f^{iv}(x) = \frac{(1+e^x)^3\cdot[e^x - 2e^{2x}] - (e^x - e^{2x})\cdot 3(1+e^x)^2\cdot e^x}{(1+e^x)^6}$$

$$= \frac{e^x - 4e^{2x} + e^{3x}}{(1+e^x)^4} \Rightarrow f^{iv}(0) = \frac{-1}{8}$$

Substituting all values in eqn (1) we get

$$\log(1+e^x) = \log 2 + x\left(\frac{1}{2}\right) + \frac{x^2}{2!}\left(\frac{1}{4}\right) + \frac{x^3}{3!}(0) + \frac{x^4}{4!}\left(\frac{-1}{8}\right) + \cdots$$

$$= \log 2 + x/2 + \frac{x^2}{8} + \frac{x^4}{192} + \cdots$$

H.P.

Ex: obtain the expansion of $\tan^{-1}x$ in ascending power of x and hence show that:

$$\sin^{-1}\left[\frac{2x}{1+x^2}\right] = 2\left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right]$$

Soln. let $y = f(x) = \tan^{-1}x$

Then by using Maclaurin's series expansion we have

$$\tan^{-1}x = y(x) = y(0) + x\,y_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \frac{x^4}{4!}y_4(0)$$

$$+ \frac{x^5}{5!}y_5(0) + \frac{x^6}{6!}y_6(0) + \frac{x^7}{7!}y_7(0) + \cdots \quad \text{—(1)}$$

Now

$$y(x) = \tan^{-1}x \quad \text{—(2)}$$

Diff. w.r.to x. we get

$$y_1(x) = \frac{1}{1+x^2} \quad \text{—(3)}$$

~~$y_1(x) =$~~ $\Rightarrow (1+x^2)y_1 = 1$

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Diff. n^{th} time we get

$$D^n\left[(1+x^2)y_1\right] = 0$$

$$\Rightarrow (1+x^2)y_{n+1} + n(2x)y_n + \frac{n(n-1)}{2!}(2)y_{n-1} = 0$$

$$\Rightarrow (1+x^2)y_{n+1} + 2nxy_n + (n^2-n)y_{n-1} = 0 \quad \text{---} (3)$$

put $x=0$, we get in eqn (1), (2) & (3) we get

(1) $\Rightarrow$ $y(0) = \tan^{-1}(0) = 0$

(2) $\Rightarrow$ $y_1(0) = \frac{1}{1+0} = 1$

(3) $\Rightarrow$ $\boxed{y_{n+1}(0) = (n-n^2)\,y_{n-1}(0)}$ ---- (4)

Now put $n = 1, 2, 3, 4, 5, 6, 7, \ldots$ in eqn (4)

$n=1 \Rightarrow$ $y_2(0) = 0$

$n=2$, $y_3(0) = (2-4)\,y_1(0) = -2$

$n=3$, $y_4(0) = (3-3^2)\,y_2(0) = 0$

$n=4$, $y_5(0) = (4-4^2)\,y_3(0) = (-12)\cdot(-2) = 24$

$n=5$, $y_6(0) = (5-5^2)\,y_4(0) = 0$

$n=6$ $y_7(0) = (6-6^2)\,y_5(0) = (-30)\cdot(24)$

$= -720$

and so on.

Now from eq^{n} (1)

$$\tan^{-1}x = y(0) + x\,y_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \frac{x^4}{4!}y_4(0) + \frac{x^5}{5!}y_5(0) + \frac{x^6}{6!}y_6(0) + \frac{x^7}{7!}y_7(0) + \cdots$$

$$= 0 + x(1) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-2) + \frac{x^4}{4!}(0) + \frac{x^5}{5!}(24) + \frac{x^6}{6!}(0) + \frac{x^7}{7!}(-720) + \cdots$$

$$\boxed{\tan^{-1}x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots}$$

~~To find~~ further

let $\theta = \tan^{-1}x \Rightarrow x = \tan\theta$

$$\therefore \sin^{-1}\left[\frac{2x}{1+x^2}\right] = \sin^{-1}\left[\frac{2\tan\theta}{1+\tan^2\theta}\right]$$

$$= \sin^{-1}[\sin 2\theta] = 2\theta$$

$$= 2\tan^{-1}x$$

$$= 2\left[x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \cdots\right]$$

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Ex: If $y = \sin(2\sin^{-1}x)$, prove that

$$(1-x^2)y_{n+2} - (2n+1)x y_{n+1} + (4-n^2)y_n = 0$$

Hence show that

$$\sin(2\sin^{-1}x) = 2x - x^3 + \frac{x^5}{4} - \cdots$$

Soln: ⇒ let $y = \sin(2\sin^{-1}x)$ ——— (1)

Diff w.r.to x. we get

$$y_1 = \cos(2\sin^{-1}x) \cdot \frac{2}{\sqrt{1-x^2}}$$

$$\Rightarrow \sqrt{1-x^2}\, y_1 = 2\cos(2\sin^{-1}x) \quad \text{——— (2)}$$

Squaring on both side we get

$$(1-x^2)y_1^2 = 4\cos^2(2\sin^{-1}x)$$
$$= 4(1 - \sin^2(2\sin^{-1}x))$$
$$= 4(1-y^2)$$

again Diff. w.r.to x we get

$$(1-x^2)2y_1y_2 + (-2x)y_1^2 = 4(0 - 2yy_1)$$

$$\Rightarrow (1-x^2)y_2 - xy_1 + 4y = 0 \quad \text{——— (3)}$$

Diff. nth times. w.r.to. x. we get

$$D^n[(1-x^2)y_2] - D^n[xy_1] + 4D^n[y] = 0$$

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using leibnitz thm. we get

$$\left[(1-x^2)y_{n+2} + n(-2x)y_{n+1} + \frac{n(n-1)}{2!}(-2)\right] - \left[x y_{n+1} + n(1)y_n\right] + y_n + 4y_n = 0$$

$$\Rightarrow \boxed{(1-x^2)y_{n+2} - (2n+1)x y_{n+1} + (4-n^2)y_n = 0} \quad \text{—(4)}$$

Now: put $x=0$ in eqn (1) (2) (3) & (4), we get

(1) $\Rightarrow$ $y(0) = \sin(2\sin^{-1}0) = 0$

(2) $\Rightarrow$ $y_1(0) = 2\cos(2\sin^{-1}0) = 2\cos(0) = 2$

(3) $\Rightarrow$ $y_2(0) = -4y(0) = 0$

(4) $\Rightarrow$ $\boxed{y_{n+2}(0) = (n^2-4)y_n(0)}$ —(5)

Now put $n = 1, 2, 3, 4, \ldots$ we get

for $n=1$: $y_3(0) = (1^2-4)y_1(0) = (-3)(2) = -6$

$n=2$ $y_4(0) = (2^2-4)y_2(0) = 0$

$n=3$ $y_5(0) = (3^2-4)\cdot y_3(0) = (5)\cdot(-6) = -30$

Now using Maclaurin's series thm.

$$\sin(2\sin^{-1}x) = y(x)$$

$$= y(0) + x y_1(0) + \frac{x^2}{2!}y_2(0) + \frac{x^3}{3!}y_3(0) + \frac{x^4}{4!}y_4(0) + \frac{x^5}{5!}y_5(0) + \ldots$$

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$$\Rightarrow \sin(2\sin^{-1}x) = 0 + x(2) + \frac{x^2}{2!}(0) + \frac{x^3}{3!}(-6) + \frac{x^4}{4!}(0) + \frac{x^5}{5!}(-30) + \cdots$$

$$\boxed{\sin(2\sin^{-1}x) = 2x - x^3 + \frac{1}{4}x^5 + \cdots}$$

H.W

(1) If $y = e^{m\sin^{-1}x}$ then find $y_n(0)$.

Hence show that

$$e^{m\sin^{-1}x} = 1 + mx + m^2\frac{x^2}{2!} + m^2(1+m^2)\frac{x^3}{3!} + \cdots$$

(2) If $y = \dfrac{\sin^{-1}x}{\sqrt{1-x^2}}$, show that

$$y_{n+2}(0) = (n+1)^2 y_n(0).$$

Hence prove that

$$\frac{\sin^{-1}x}{\sqrt{1-x^2}} = x + \frac{2}{3}x^3 + \frac{8}{15}x^5 + \cdots$$

(3) Show that

(i) $\sin x = x - x^3/3! + x^5/5! - \frac{x^7}{7!} + \cdots$

(ii) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots$

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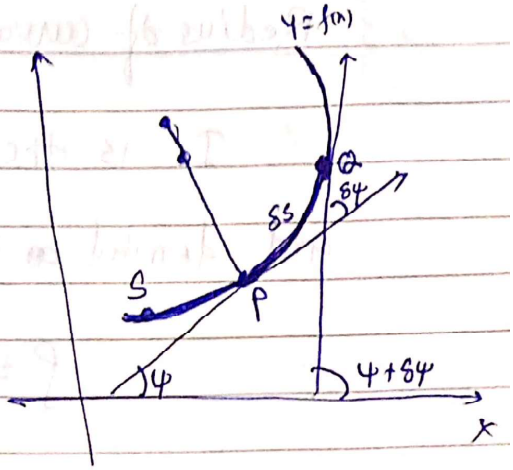
Redius of curvature :

Consider a curve $y = f(x)$ in xy-plane

and let S be the fixed point on it.



let P & Q be any two point on the curve $y = f(x)$

let $\psi \Rightarrow$ angle made by tangent at P with x-axis

$\psi + \delta\psi \Rightarrow$ angle made by tangent at Q with x-axis.

Here $\delta\psi \Rightarrow$ angle of contigence (angle betn two tangents) of arc PQ

& $\delta s \Rightarrow$ length of arc PQ

Then Average Curvature of arc $PQ = \dfrac{\delta\psi}{\delta s}$

Curvature :

Curvature of a curve $y = f(x)$, at pt P. is the limiting value of the average curvature as pt $Q \to P$.

ie. Curvature at $P = \lim_{Q \to P}$ (Average curvature)

$\Rightarrow$ $= \lim_{\delta s \to 0} \dfrac{\delta\psi}{\delta s} =$

$= \dfrac{d\psi}{ds}$

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Radius of curvature

It is the reciprocal of curvature at pt. P. and denoted as by

$$\rho = \frac{1}{\left(\frac{d\psi}{ds}\right)} = \frac{ds}{d\psi}$$

★ For cartesian curve

at pt (x, y)

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

★ For parametric curve :

Let $x = f(\theta)$, $y = g(\theta)$

Then

$$\boxed{\rho = \frac{(x_1^2 + y_1^2)^{3/2}}{x_1 y_2 - y_1 x_2}}$$

where, $x_1 \equiv \frac{dx}{d\theta}$, $y_1 = \frac{dy}{d\theta}$

$x_2 = \frac{d^2x}{d\theta^2}$, $y_2 = \frac{d^2y}{d\theta^2}$

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★ For polar curve

let $r = f(\theta)$ be the polar curve.

Then

$$\rho = \frac{[r^2 + (r_1)^2]^{3/2}}{r^2 + 2(r_1)^2 - r \cdot r_2}$$

OR when

$$\boxed{\rho = \frac{\left[r^2 + (dr/d\theta)^2\right]^{3/2}}{r^2 + 2\left(\frac{dr}{d\theta}\right)^2 - r\frac{d^2r}{d\theta^2}}}$$

★ Centre of curvature :

let $(\bar{x}, \bar{y})$ be the centre of curvature for curve $y = f(x)$

Here $$\bar{x} = x - \left[\frac{y_1(1+y_1^2)}{y_2}\right] = x - \left(\frac{\frac{dy}{dx}\left(1+\left(\frac{dy}{dx}\right)^2\right)}{\frac{d^2y}{dx^2}}\right)$$

$$\bar{y} = y + \left[\frac{(1+y_1^2)}{y_2}\right] = y + \left(\frac{1+\left(\frac{dy}{dx}\right)^2}{\frac{d^2y}{dx^2}}\right)$$

★ Circle of curvature :

an eqn $$\boxed{(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2}$$

is called circle of curvature with centre at $(\bar{x}, \bar{y})$ and radius equal to ρ.

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Ex: For the curve $(a+x)y = ax$

If ρ is radius of curvature at any pt (x,y). Show that

$$\left(\frac{2\rho}{a}\right)^{2/3} = \left(\frac{x}{y}\right)^2 + \left(\frac{y}{x}\right)^2$$

Soln: Given curve is

$$(a+x)y = ax$$

$$\Rightarrow y = \frac{ax}{(a+x)} \quad \text{---(1)}$$

Diff w.r.to x. we get

$$y_1 = \frac{(a+x)\cdot(a) - (1)\,ax}{(a+x)^2} = \frac{a^2}{(a+x)^2}$$

again diff w.r.to x. we get

$$y_2 = \frac{-2a^2}{(a+x)^3}$$

For the given curve

$$y_1 = \frac{a^2}{(x+y)^2} = \frac{a^2}{(ax/y)^2} = \frac{y^2}{x^2}$$

$$y_2 = \frac{-2a^2}{(a+x)^3} = \frac{-2a^2}{(ax/y)^3} = \frac{-2y^3}{a\cdot x^3}$$

$y_2 = \frac{d}{dx}\left(\frac{y^2}{x^2}\right)$

$= \frac{x^2 \cdot 2yy_1 - 2xy^2}{x^4}$

$= 2\left[\frac{x^2 y^3/x^2 - y^2 x}{x^4}\right]$

$= 2\{$

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∴ Radius of curvature

$$\rho = \frac{[1+(y_1)^2]^{3/2}}{y_2}$$

$$= \frac{\left(1+(y^2/x^2)^2\right)^{3/2}}{\left(-\frac{2y^3}{ax^3}\right)}$$ (ignore +ve sign)

$$\Rightarrow \left(\frac{2}{a}\rho\right) = \frac{\left(1+y^4/x^4\right)^{3/2}}{\left(\frac{y^3}{x^3}\right)}$$

Taking 2/3 power on both side

$$\left(\frac{2}{a}\rho\right)^{2/3} = \frac{\left(\left[1+y^4/x^4\right]^{3/2}\right)^{2/3}}{\left(\frac{y}{x}\right)^{3\cdot 2/3}}$$

$$\Rightarrow \left(\frac{2}{a}\rho\right)^{2/3} = \frac{1+y^4/x^4}{(y/x)^2} = \frac{(x^4+y^4)/x^4}{y^2/x^2}$$

$$= \frac{x^4+y^4}{x^2\cdot y^2}$$

$$= \frac{x^2}{y^2} + \frac{y^2}{x^2}$$

$$\Rightarrow \boxed{\left(\frac{2}{a}\rho\right)^{2/3} = \left(\frac{x}{y}\right)^2 + \left(\frac{y}{x}\right)^2}$$

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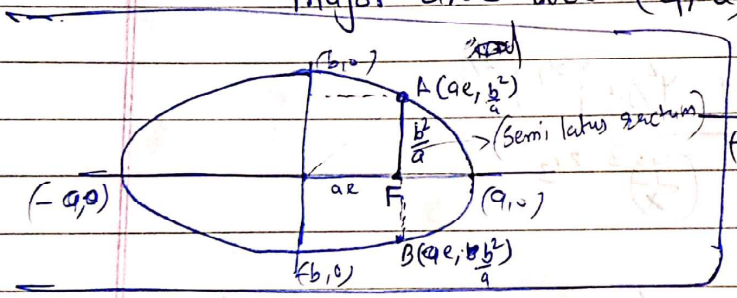
Ex. In the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

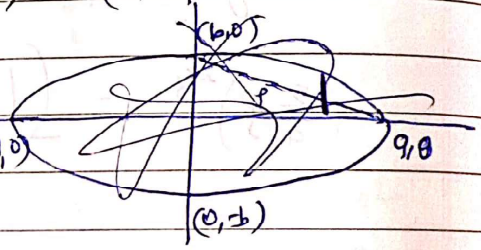
Show that the radius of curvature at an end of the major axis is equal to the semi latus rectum of the ellipse

i.e. $\boxed{\rho = \frac{b^2}{a}}$

Soln. let $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ — (1)

be the ellipse. then end point of its major axis are $(a, 0)$, $(-a, 0)$





Diff eqn (1) w.r.to x. we get

$$\frac{2x}{a^2} + \frac{2y}{b^2}\frac{dy}{dx} = 0$$

$$\Rightarrow y_1 = \frac{dy}{dx} = \frac{-b^2}{a^2}\left(\frac{x}{y}\right)$$

again diff we get

$$y_2 = \frac{d^2y}{dx^2} = \frac{-b^2}{a^2}\left[\frac{y(1) - x\,dy/dx}{y^2}\right]$$

$$= \frac{-b^2}{a^2}\left[\frac{y - x\left(-b^2/a^2 \cdot (x/y)\right)}{y^2}\right]$$

$$= \frac{-b^2}{y^2}\left(\frac{a^2y^2 + b^2x^2}{a^2y^3}\right)$$

$$= \frac{-b^2}{a^4y^3} \cdot (a^2b^2)$$

$\because \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\Rightarrow a^2y^2 + x^2b^2 = a^2b^2$

$$\Rightarrow y_2 = \frac{d^2y}{dx^2} = \frac{-b^4}{a^2y^3}.$$

Thus radius of curvature at any pt (x,y) is

$$\rho_{\text{at }(x,y)} = \frac{[1 + (y_1)^2]^{3/2}}{y_2}$$

$$= \frac{\left[1 + \left(\frac{-b^2}{a^2}(x/y)\right)^2\right]^{3/2}}{\frac{-b^4}{a^2y^3}}$$

$$= \frac{(a^4y^2 + b^4x^2)^{3/2}}{a^6y^3} \cdot \left(\frac{-a^2y^3}{b^4}\right)$$

ignoring -ve sin

$$= \frac{(a^4y^2 + b^4x^2)^{3/2}}{a^4b^4} \cdot \rho$$

at pt. $(a,0)$. (end pt of major axis of an ellipse)

$$\rho_{\text{at }(a,0)} = \frac{[a^4(0) + b^4(a)^2]^{3/2}}{a^4b^4} = \frac{b^6a^3}{a^4b^4}$$

$$= \frac{b^2}{a}$$

= Semi latus rectum.

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Ex: If $x = a\cos^4\theta$, $y = a\sin^4\theta$, find the curvature at $\theta = \pi/6$

Soln: let $x = a\cos^4\theta$, $y = a\sin^4\theta$

$$\Rightarrow x_1 = \frac{dx}{d\theta} = -4a\cos^3\theta\cdot\sin\theta \qquad y_1 = \frac{dy}{dt} = 4a\sin^3\theta\cdot\cos\theta$$

Thus

$$\frac{dy}{dx} = \frac{dy}{d\theta}\cdot\frac{d\theta}{dx}$$

$$= 4a\sin^3\theta\cdot\cos\theta\cdot\frac{1}{-4a\cos^3\theta\cdot\sin\theta}$$

$$= -\frac{\sin^2\theta}{\cos^2\theta}$$

$$= -\tan^2\theta$$

Now, $$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right)$$

$$= \frac{d}{d\theta}\left(\frac{dy}{dx}\right)\cdot\frac{d\theta}{dx}$$

$$= \left[\frac{d}{d\theta}\left(-\tan^2\theta\right)\cdot\right]\times\frac{1}{-4a\cos^3\theta\cdot\sin\theta}$$

$$= \frac{-2\tan\theta\cdot\sec^2\theta}{-4a\cos^3\theta\cdot\sin\theta}$$

$$\frac{d^2y}{dx^2} = \frac{1}{2a}\frac{1}{\cos^6\theta}$$

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Now. Radius of curvature at any pt. (x, y) is given by

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\frac{d^2y}{dx^2}}$$

$$= \frac{\left[1 + (-\tan^2\theta)^2\right]^{3/2}}{\frac{1}{2a} \quad \frac{1}{\cos^6\theta}} = 2a\left(1 + \tan^2\theta\right)^{3/2} \cos^6\theta$$

put $\theta = \pi/6$

$$\therefore \rho = 2a\left(1 + \tan^2 \pi/6\right)^{3/2} \cdot \left[\cos(\pi/6)\right]^6$$

$$= 2a\left[1 + (1/\sqrt{3})^2\right]^{3/2} \left(\sqrt{3}/2\right)^6$$

$$= 2a\left[10/9\right]^{3/2} \left(3/4\right)^3$$

$$= \frac{2a\,(10)^{3/2}}{(9)^{3/2}} \cdot \frac{3^3}{4^3} = \frac{2a\,(10)^{3/2} \cdot 27}{64 \cdot (27)}$$

$$= \frac{a\,(10)^{3/2}}{32}$$

$$\therefore \rho = \frac{(10)^{3/2} \cdot a}{32}$$

Thus

$$\text{Curvature} = \frac{1}{\rho}$$

$$= \frac{32}{a\,(10)^{3/2}}$$

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Ex. Prove that the circle of curvature of the curve $x^3+y^3=3axy$ at the point $\left(\frac{3a}{2}, \frac{3a}{2}\right)$ is

$$\left(x-\frac{21a}{16}\right)^2+\left(y-\frac{21a}{16}\right)^2=\left(\frac{3a}{8\sqrt{2}}\right)^2$$

Soln: Given curve

$$x^3+y^3=3xya \quad \text{---} (1)$$

Diff. w.r.to x. we get

$$3x^2+3y^2\cdot y_1=3ax\,y_1+3ay$$

$$\Rightarrow (3y^2-3ax)\,y_1=3ay-3x^2$$

$$\Rightarrow (y^2-ax)\,y_1=(ay-x^2)$$

$$\Rightarrow y_1=\left(\frac{ay-x^2}{y^2-ax}\right)$$

again diff w.r.to x. we get

$$\frac{d^2y}{dx^2}=y_2=\frac{(y^2-ax)\cdot(ay_1-2x)-(ay-x^2)\cdot(2yy_1-a)}{(y^2-ax)^2}$$

Now at pt. $\left(\frac{3a}{2}, \frac{3a}{2}\right)=(x,y)$. we get

$$y_1=\frac{dy}{dx}=\frac{a(3a/2)-(3a/2)^2}{(3a/2)^2-a(3a/2)}=-\frac{\left[\left(\frac{3a}{2}\right)^2-\frac{3a^2}{2}\right]}{\left(\left(\frac{3a}{2}\right)^2-\frac{3a^2}{2}\right)}$$

$$=-1$$

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and $y_2 = \frac{d^2y}{dx^2} = \left[\left(\left(\frac{3a}{2}\right)^2 - \frac{3a^2}{2}\right)\left(a\frac{(-1)}{}^{?} - 2\frac{3a}{2}\right) - \left[a\frac{3a}{2} - \left(\frac{3a}{2}\right)^2\right]\left(2\left(\frac{3a}{2}\right)\cdot(-1) - a\right)\right]$

$$\left(\left(\frac{3a}{2}\right)^2 - 3\frac{a^2}{2}\right)^2$$

$$= \frac{-32}{3a}$$

Thus radius of curvature at pt $(3a/2, 3a/2)$ is

$$\rho = \frac{\left[1+(y_1)^2\right]^{3/2}}{y_2} = \frac{\left[1+(-1)^2\right]^{3/2}}{\left(\frac{-32}{3a}\right)}$$

$$\boxed{\rho = \frac{3a\,(2^{3/2})}{32}} \qquad (-ve \text{ ignored})$$

Now To find ... Centre of curvature

$$\bar{x} = x - \left[\frac{y_1\left(1+(y_1)^2\right)}{y_2}\right]$$

$$= \frac{3a}{2} - \left[\frac{(-1)\left(1+(-1)^2\right)}{(-32/3a)}\right] = \frac{3}{2}a - 2\cdot\frac{3a}{32}$$

$$\boxed{\bar{x} = \frac{21}{16}a}$$

$$\bar{y} = y + \frac{\left(1+(y_1)^2\right)}{y_2} = \frac{3a}{2} + \frac{\left(1+(-1)^2\right)}{(-32/3a)}$$

$$\boxed{\bar{y} = \frac{21}{16}a}$$

Now. Circle of curvature with centre at $(\frac{39}{2}, \frac{39}{2})$ and radius equal to ρ. is given by,

$$(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$$

$$\Rightarrow \left(x - \frac{219}{16}\right)^2 + \left(y - \frac{219}{16}\right)^2 = \left[\frac{2^{3/2} \cdot 39}{32}\right]^2$$

$$= \left(\frac{39 \cdot \sqrt{2}}{16}\right)^2$$

$$\boxed{\left(x - \frac{219}{16}\right)^2 + \left(y - \frac{219}{16}\right)^2 = \left(\frac{39}{8\sqrt{2}}\right)^2}$$

Ex: Show that circle of curvature at the origin for the curve

$$x+y = ax^2 + by^2 + cx^3 \text{ is}$$

$$(a+b)(x^2+y^2) = 2(x+y).$$

Soln. Given curve is

$$x+y = ax^2 + by^2 + cx^3 \quad \text{---(1)}$$

Diff. w.r.to. x, we get

$$1 + \frac{dy}{dx} = 2ax + 2by \cdot \frac{dy}{dx} + 3cx^2.$$

$$\therefore \frac{dy}{dx}(1-2by) = 2ax + 3cx^2 - 1$$

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$$\Rightarrow \frac{dy}{dx} = \left(\frac{2ax + 3(x^2 - 1}{1 - 2by}\right).$$

again diff w.r.to x. we get

$$\frac{d^2y}{dx^2} = \frac{(1-2by)\cdot[2a + 6cx] - [2ax + 3(x^2 - 1]\cdot(-2b\frac{dy}{dx})}{(1-2by)^2}$$

at the origin i.e. pt (0,0), we have

$$\frac{dy}{dx} = \frac{2a(0) + 3c(0) - 1}{1 - 2b(0)} = -1$$

&

$$\frac{d^2y}{dx^2} = 2a + 2b$$

Thus : Radius of curvature at origin.

$$\rho = \frac{\left[1 + \left(\frac{dy}{dx}\right)^2\right]^{3/2}}{\left(\frac{d^2y}{dx^2}\right)} = \frac{[1 + (-1)^2]^{3/2}}{2a + 2b} = \frac{\sqrt{2}}{a+b}.$$

Now. Centre of curvature $(\bar{x}, \bar{y})$ given by: (at origin)

$$\bar{x} = x - \left[\frac{y_1(1 + y_1^2)}{y_2}\right] = 0 - \frac{(-1)(1 + (-1)^2)}{2a + 2b}$$

$$= \frac{1}{a+b}$$

$$\bar{y} = y + \frac{[1 + (y_1)^2]}{y_2} = 0 + \frac{1 + (-1)^2}{2a + 2b} = \frac{1}{a+b}$$

Then. Centre of curvature is.

$$(\bar{x}, \bar{y}) = \left(\frac{1}{a+b}, \frac{1}{a+b}\right)$$

Now, Circle of curvature is given by

$$(x-\bar{x})^2 + (y-\bar{y})^2 = \rho^2$$

$$\Rightarrow \left(x - \frac{1}{a+b}\right)^2 + \left(y - \frac{1}{a+b}\right)^2 = \left(\frac{\sqrt{2}}{a+b}\right)^2$$

$$\Rightarrow x^2+y^2 - \frac{2x}{a+b} - \frac{2y}{a+b} = 0$$

$$\Rightarrow (a+b)(x^2+y^2) - (2x+2y) = 0$$

$$\Rightarrow \boxed{(a+b)(x^2+y^2) = 2x+2y}$$

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Ex: Find the radius of curvature at any pt on the cardioid $r = a(1+\cos\theta)$ and prove that $\frac{\rho^2}{r} = \text{const.}$

Soln: Let

$$r = a(1+\cos\theta)$$

$$\Rightarrow r_1 = -a\sin\theta$$

$$r_2 = -a\cos\theta$$

$$\therefore \rho = \frac{(r^2 + r_1^2)^{3/2}}{r^2 + 2r_1^2 - rr_2} = \frac{[a^2(1+\cos\theta)^2 + a^2\sin^2\theta]^{3/2}}{a^2(1+\cos\theta)^2 + 2a^2\sin^2\theta + a^2\cos\theta(1+\cos\theta)}$$

$$= \frac{(2+2\cos\theta)^{3/2}}{3(1+\cos\theta)}$$

$$\boxed{\rho = \frac{a}{3} r^{3/2}(1+\cos\theta)^{3/2}}$$

$$= \frac{8a}{9}$$

$$\therefore \rho^2 = \frac{4}{9} \cdot 2a^2 (1+\cos\theta)$$

$$= \frac{8a^2}{9}[a(1+\cos\theta)]$$

$$= \frac{8a}{9} r$$

$$\therefore \boxed{\frac{\rho^2}{r} = \frac{8a}{9} = \text{const}}$$